

Making AI Drafts Count: A Quality Threshold in Audio Description Workflows

Lana Do*
Northeastern University
United States
do.ng@northeastern.edu

Shasta Ihorn
San Francisco State University
United States
sihorn@sfsu.edu

Charity M. Pitcher-Cooper
The Smith-Kettlewell Eye Research
Institute
United States
cpc@ski.org

Sanjay Mirani
San Francisco State University
United States
mirisanjanjay417@gmail.com

Gio Jung
San Francisco State University
United States
gjung1@sfsu.edu

Hyunjoo Shim
San Francisco State University
United States
shim.hy@northeastern.edu

Zhenzhen Qin
Northeastern University
United States
qin.zhen@northeastern.edu

Kien T. Nguyen
Northeastern University
United States
nguyen.trungk@northeastern.edu

Vassilis Athitsos
University of Texas at Arlington
United States
athitsos@uta.edu

Ilmi Yoon
Northeastern University
United States
yoon.i@northeastern.edu

Abstract

Audio description (AD) narrates visual elements in video for blind and low-vision audiences. Recent work has shown that giving novice describers an AI-generated draft to start from helps produce higher-quality AD and lowers the barrier to entry. What remains an open question is how draft quality shapes the editing process. We investigate this through GenAD, an AD generation pipeline that incorporates accessibility guidelines and contextual video information, and RefineAD, an editing interface for human revisions. Human-AI contributions are measured across text, timing, and delivery. In a within-subjects study, we compared authoring from scratch against editing AI drafts of varying quality. GenAD drafts cut completion time by more than half and significantly reduced cognitive load. In contrast, baseline drafts generated from simple, unguided prompts offered only modest benefits, pointing to a minimum quality threshold for effectiveness. Qualitative findings suggest this threshold is content-dependent; as visual complexity increases, so does the quality needed from AI drafts. We propose this as a design principle: effective AI assistance should clear a

quality threshold suited to the target content, rather than simply be present.

CCS Concepts

• **Human-centered computing** → **Accessibility technologies**.

Keywords

Audio Description; Human-AI Collaboration; Draft Quality

ACM Reference Format:

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1 Introduction

Audio description (AD) conveys essential visual information through spoken narration, making video content accessible to blind and low-vision (BLV) audiences. Professional AD is costly and time-consuming [6], leaving the growing volume of user-generated and short-form video content largely undescribed [36]. Multimodal Large Language Models (MLLMs) have opened a path to generating AD at scale [12, 47, 49], but AI-produced descriptions still suffer from hallucinations, verbosity, and lack of contextual coherence [20, 27]. Human oversight remains necessary. In practice, audio description quality stewardship falls to volunteer describers, the primary contributors bringing accessible content to BLV users. Designing effective editing environments for volunteer authors is

*Corresponding author.

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not straightforward: AD editing requires visual judgment, temporal reasoning, and domain knowledge of accessibility conventions [31, 32, 35].

Prior work demonstrates that human-AI workflows for AD are viable and can lower the barrier to entry for novice describers. Yuksel et al. showed that starting from an AI-generated draft was faster and easier for first-time volunteers on YouDescribe than working without AI assistance, even when describer-rated draft accuracy was only neutral [52]. Cheema et al. extended this with collaborative co-authoring that reduces repetitive effort and eases cognitive load [8]. However, neither study systematically varied the quality of the AI drafts to test their specific impact on performance.

In practice, MLLM output varies substantially with prompting strategy [12, 42]. While prior work established that prompting MLLMs with professional AD conventions improves description quality for BLV users [9, 28], how that variation affects the downstream editing experience has not been studied. Evidence from machine translation (MT) post-editing suggests that output quality matters: higher-quality output reduces editor workload, while low-quality output can be counterproductive—for example, editors who become anchored to flawed translations may spend more effort than if they had translated from scratch [17, 24]. Whether a similar quality threshold exists for AD—specifically, the point at which the effort of correcting AI errors outweighs the benefit of the draft—remains an open question.

We address this through the co-design of two components: **GenAD**, an accessibility-informed generation pipeline that incorporates professional AD guidelines [1, 33, 50] and contextual video information to produce structured drafts; and **RefineAD**, an editing interface organized around content editing, temporal alignment, voice delivery, and workflow support. To measure human contribution in this hybrid authoring process, we introduce a *Multi-Dimensional Contribution Index* (MDCI) that decomposes each editor’s modifications relative to the AI-generated baseline across four dimensions: text, timing, playback, and voice. RefineAD also supports a collaborative mode in which multiple contributors can build incrementally on each other’s work. Both components were designed in consultation with accessibility experts, including an experienced audio describer who trains novice volunteers and a professional blind quality controller.

To evaluate whether structured drafts, driven by AD guidelines and contextual grounding, are necessary for novices to author descriptions productively, we conducted a within-subjects study with 30 participants across three randomized and anonymized conditions: *from scratch* (no AI draft), *Baseline* (unguided AI drafts), and *GenAD* (guideline-informed and contextually grounded). The results reveal a threshold effect: while GenAD drafts significantly reduced both completion time and cognitive load, Baseline drafts offered a modest time reduction and did not meaningfully decrease cognitive load compared to working from scratch.

In summary, this paper contributes:

- (1) **A design principle for the usability threshold in human-AI authoring:** Empirical results indicate that AI drafts must reach a sufficient quality threshold to meaningfully reduce cognitive load; below it, unguided drafts offer only modest speed gains. Qualitative evidence suggests this threshold varies with

content complexity, with domain-specific material requiring higher draft quality.

- (2) **GenAD and RefineAD**, a co-designed system for guideline-informed AD authoring featuring a Multi-Dimensional Contribution Index (MDCI) to track human and AI input across content, timing, and delivery.

2 Related Work

2.1 AI-Driven and Participatory AD Systems

The scarcity of professional AD and the growth of online video content have motivated automated generation. Wang et al. built Tiresias, combining audiovisual inconsistency detection, dense video captioning, and GPT-based optimization to generate and place AD [47]. LLM-AD used GPT-4V with scene segmentation and character tracking [12], while MMAD combined CLIP with LLaMA2 to integrate subtitles and sound cues for narrative consistency [49]. ShortScribe used BLIP-2 and GPT-4 to generate hierarchical video summaries at multiple levels of detail [44]. Most recently, VideoA11y prompted GPT-4V with 42 professional AD guidelines to generate descriptions [28]. However, automated AD alone often falls short. Systems misidentify characters or insert hallucinated details [27, 47], produce verbose or redundant descriptions [20], and score well on captioning metrics yet misalign with human judgments of AD quality [38]. AI-generated descriptions have also been shown to encode societal biases, defaulting to hegemonic assumptions about race, gender, and appearance [3]. Moreover, AD has been characterized as a creative and interpretive practice involving subjectivity and stylistic nuance, qualities that remain difficult to automate [40, 45].

Beyond automation, crowdsourcing offers another path to expanding AD coverage. YouDescribe provides a web platform where volunteers record descriptions synchronized to YouTube videos [50]. A range of tools have been developed to support AD authoring: Rescribe [35] and CrossA11y [31] reduced authoring barriers through automatic editing and cross-modal accessibility detection, respectively. OmniScribe extended AD authoring to 360° video [7], and AVscript made the process accessible to BLV creators through text-based audio-visual scripts [19].

A growing line of work explores human-in-the-loop (HITL) workflows where AI generates a draft and a human editor refines it. Yuksel et al. demonstrated that novice describers working from AI drafts on YouDescribe were significantly faster than those writing from scratch [4, 52], and Cheema et al. extended this with DescribePro, which supports iterative refinement through natural language prompting [8]. These studies establish that HITL systems are viable and can reduce the burden on volunteers.

2.2 From Output Quality to Editing Experience

Human-AI workflows depend on AI-generated drafts, but draft quality varies substantially with how the underlying model is prompted. Generic prompts tend to produce verbose, context-poor output resembling captions more than coherent AD [11, 30], while incorporating professional AD guidelines into prompts produces descriptions comparable to trained describers [28]. AutoAD-Zero further demonstrated that prompt design alone can reach competitive performance without fine-tuning [48].

However, these studies evaluate the impact of prompting strategies via BLV end-user ratings of output quality; whether these gains improve the editing experience of describers remains unknown. This question is critical for volunteer-driven platforms like YouDescribe, which rely on novice contributors. Prior work in adjacent domains shows that draft quality directly shapes editing effort. In machine translation post-editing, higher-quality outputs reduce workload, while low-quality outputs can be counterproductive, sometimes requiring more effort than translating from scratch [17, 24]. Error type further modulates effort, indicating that aggregate quality is an insufficient predictor [13]. Similarly, in AI-assisted writing, low-level scaffolding offers little benefit over unassisted writing, whereas high-level scaffolding improves quality but can reduce perceived ownership and satisfaction [14].

These findings suggest a quality threshold below which AI assistance hinders rather than helps. Whether such a threshold exists for audio description (AD) authoring remains untested. AD editing introduces distinct constraints—describers must interpret visual scenes, align text to tight temporal windows, and adhere to accessibility conventions—making direct transfer from prior domains uncertain.

3 System

GenAD and RefineAD were designed for community-driven AD platforms where amateur volunteers describe videos without professional tools. Retention on such platforms is limited by the cognitive demands of authoring, motivating AI generated drafts.

3.1 GenAD: AD Generation Pipeline

GenAD generates audio descriptions for YouTube videos on demand, producing drafts that volunteers can preview and refine in RefineAD. Consistent with DCMF guidelines recommending that describers work with familiar content [1], the system allows users to request any YouTube video, enabling them to begin with material they are familiar with. The user’s request triggers a four-stage pipeline: video processing, scene-level generation, description optimization, and text-to-speech (TTS) conversion (Figure 1, A–D). Each stage is informed by iterative feedback from a professional audio describer who trains novice volunteers. The output is a complete set of descriptions synchronized with the requested video.

3.1.1 Video Processing (A). The pipeline downloads the video and metadata via yt-dlp [51], extracts frames with ffmpeg [43], and segments scenes using cosine similarity of OpenCLIP frame embeddings [41].

3.1.2 Scene-Level Generation (B). Early prototyping revealed that prompting the MLLM (GPT-4o [34]) with minimal instruction (e.g., “describe the scene”) produced caption-like outputs rather than coherent descriptions. These were verbose, lacked narrative continuity, and repeated information inferable from dialogue. Our consultants recognized these as common mistakes by novice describers who “try to do too much.” To address this, GenAD integrates three inputs (Figure 1, B1): professional guidelines that emphasize accurate and descriptive yet concise narration [1, 33, 50], the scene’s audio transcript, and accumulated descriptions from prior scenes (full prompt in Appendix A). Each scene produces structured output (Figure 1,

B2) with precise timestamps, two event types, *Text-on-Screen* and *Visual*, and track assignment (*Inline* or *Extended*). Inline narration fits within natural pauses in dialogue, and extended narration briefly pauses playback to deliver additional detail [50].

3.1.3 Description Optimization (C). Based on our consultants’ feedback to minimize playback interruptions, GenAD prioritizes inline over extended descriptions. An optimization pass condenses current extended clips to fit within dialogue gaps, using retry prompts that progressively shorten text while preserving critical details. If inline delivery remains unfeasible, the system applies a filtering prompt to retain only extended clips that describe necessary information—content not inferable from the audio or prior scenes (Figure 1, C).

3.1.4 Text-to-Speech (D). Optimized descriptions are synthesized into spoken audio using Google TTS [16], with distinct voices for Visual and Text-on-Screen events (Figure 1, D). This was informed by our consultants’ feedback that BLV users find the repeated phrase “text on screen” fatiguing; a distinct voice lets listeners identify the event type without explicit labeling as well as supporting the broader goal of keeping descriptions concise.

Finally, the audio clips are synchronized with the video for preview and refinement, with inline and extended narration color-coded on the timeline (yellow and purple, respectively), and audio ducking controls balance narration volume with the original audio.

3.1.5 Effect of Prompting Strategy on Output Quality. We briefly illustrate qualitative differences between the two generation approaches in Figure 2. The unguided condition uses only a simple prompt (“describe the scene”) without guidelines or context, producing generic outputs like “a woman stands in the desert.” With GenAD’s full pipeline, the same scene identifies the character by name (Figure 2a), and the optimization pass condenses overlong drafts into inline descriptions that fit natural dialogue gaps (Figure 2b). A quantitative comparison across all study videos is presented in Section 4.

3.2 RefineAD: Human-AI AD Authoring

Once GenAD completes processing, the original requester is notified via email and linked directly to the editing interface (Figure 1, E). The AI draft is also visible to all platform users, if another editor wants to describe the same video, they can begin editing from the preview interface (Appendix B) without submitting a duplicate request. Each editor’s work is saved to their account and can be revisited and edited at any time. All user-facing components, including viewing and editing, were developed and tested with multiple screen readers (JAWS, NVDA, and VoiceOver) and confirmed fully navigable by our blind consultant. RefineAD preserves the layout of the original freestyle editor (Appendix C), with preloaded descriptions as the only visible change. GenAD’s output serves as editable scaffolding, time-stamped, typed descriptions that editors can accept, modify, or replace. The interface is organized around four aspects of AD editing (Figure 3).

3.2.1 Workflow Support. AD authoring requires describers to simultaneously attend to visual content, dialogue timing, and description quality. To manage this complexity, the interface provides a notes panel for drafting observations during review and a dialog

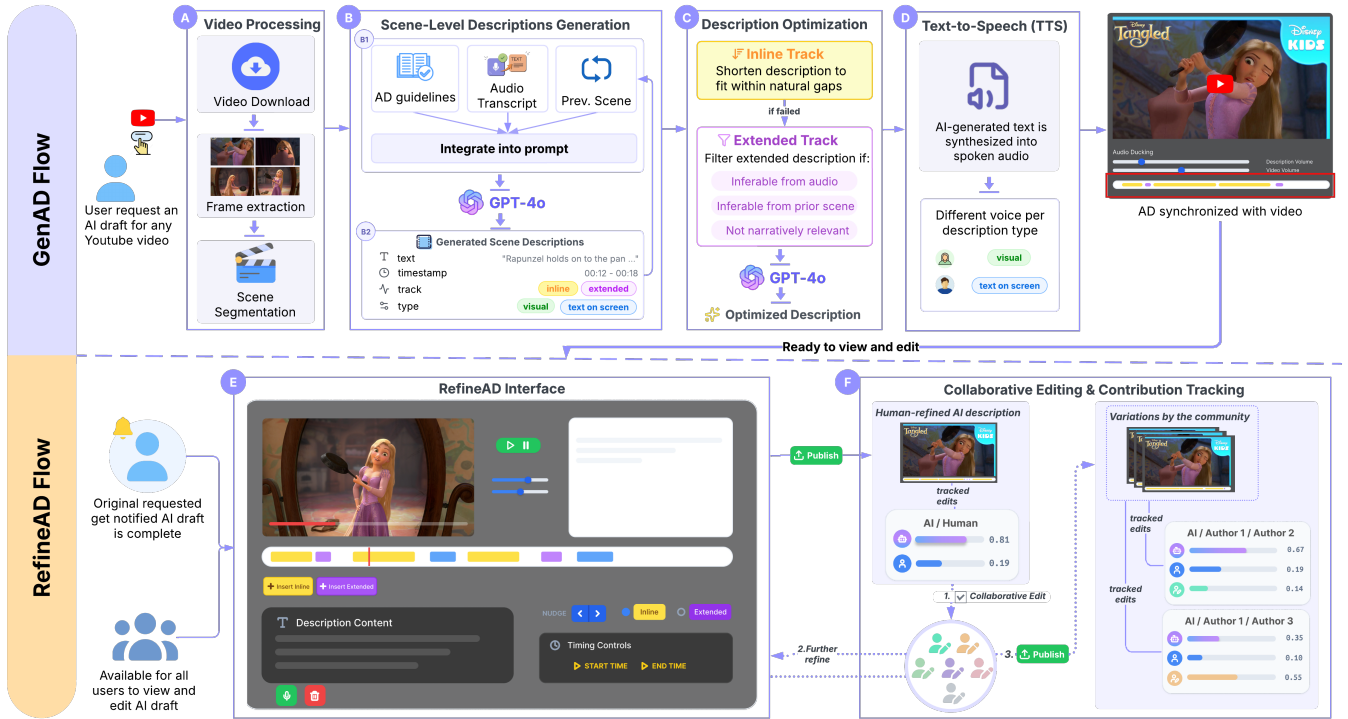
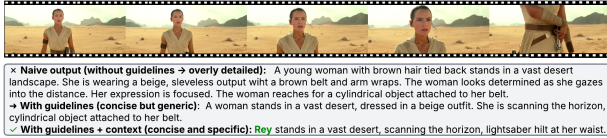
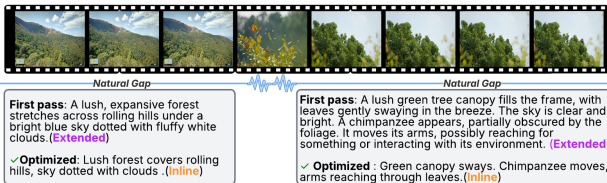


Figure 1: Overview of the GenAD-RefineAD pipeline. GenAD processes a YouTube video through four stages: (A) video processing, (B) scene-level description generation incorporating AD guidelines, audio transcript, and prior scene context, (C) description optimization that prioritizes inline delivery and filters extended descriptions, and (D) text-to-speech synthesis with distinct voices per event type. The output feeds into (E) the RefineAD editing interface, where editors can preview, edit, and publish descriptions, and (F) collaborative editing with multi-author contribution tracking.



(a) Star Wars Trailer – progression from simple prompting to guideline-informed and context-aware prompting.



(b) Jane Goodall clip – optimization condenses extended description into descriptions that fit natural pauses.

Figure 2: Qualitative differences between Baseline and GenAD output, shown on videos outside the study corpus.

timeline that visualizes all clips alongside dialogue regions. As the video plays with descriptions synchronized, the active clip is highlighted, letting users evaluate descriptions in context rather than in isolation.

3.2.2 Content Authoring. Users can reshape both the content and structure of the AI draft: inserting new descriptions, editing text, deleting unnecessary clips, or toggling between inline and extended

track types. Each clip is pre-labeled by event type (Visual or Text-on-Screen). This design positions AI output as a starting point rather than a finished product, giving users the ability to reshape descriptions based on their own interpretation.

3.2.3 Temporal Alignment. Descriptions need to avoid overlapping dialogue and align with the visual events they narrate. Fine-grained $\pm 0.25s$ nudge controls support incremental adjustment, while direct start and end time editing enables precise placement. The dialog timeline provides visual feedback on whether descriptions fall within or outside dialogue gaps, reducing the trial-and-error typically involved in timing AD.

3.2.4 Voice and Delivery. Each clip includes an AI text-to-speech track. After editing text, editors can regenerate the AI voice to reflect their changes or record their own voiceover to replace it entirely. Research shows that BLV audiences often prefer human narration for its clarity and expressiveness, particularly in dramatic or fast-paced content [15, 46]. This supports a range of workflows, from quick edits using AI voice to fully human-narrated descriptions.

3.3 Contribution Tracking

When ready to publish, describers can opt into collaborative editing, allowing others to build on their draft. The system tracks contributions across AI and human collaborators, and when collaborative

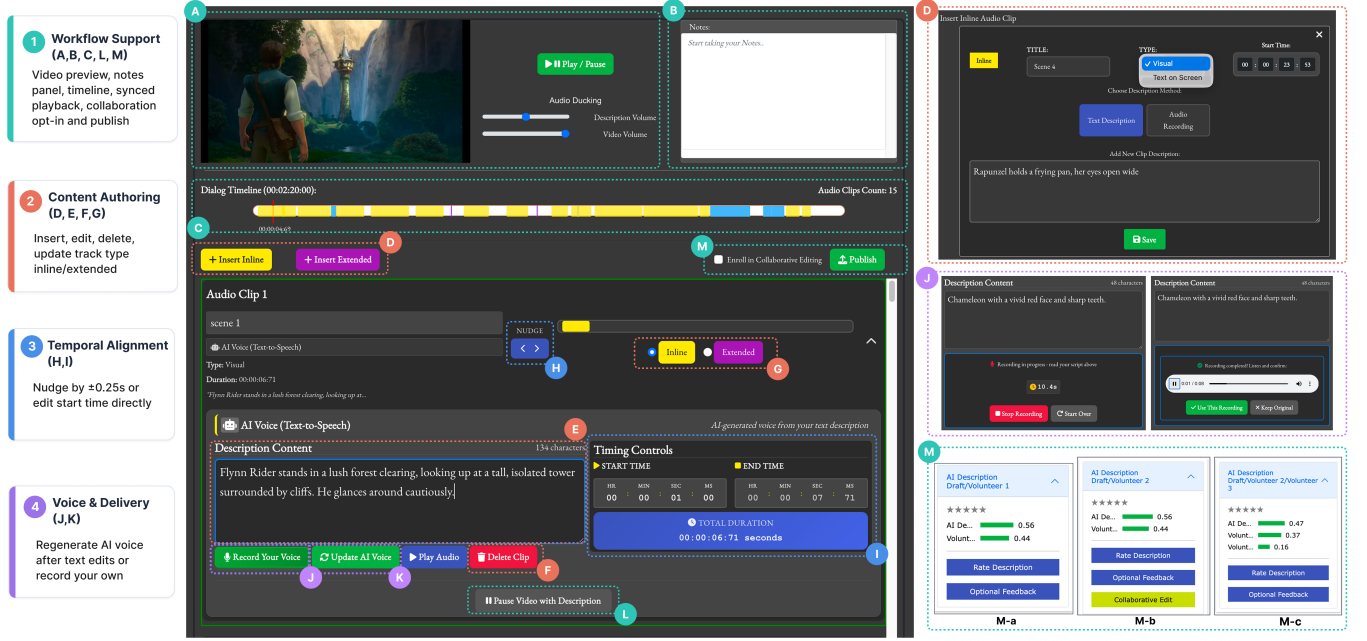


Figure 3: The RefineAD editing interface. Describers preview the video with audio ducking controls (A), take notes (B), and navigate descriptions via a color-coded dialog timeline (C; yellow = inline, purple = extended, blue = dialogue). Editors can insert new inline or extended descriptions (D), edit text (E), delete clips (F), toggle track type (G), adjust timing via $\pm 0.25s$ nudge or direct entry (H, I), and record a personal voiceover (J) or regenerate the AI voice for updated text (K). Playback synchronizes with the active clip for in-context review (L). Describers publish and can opt into collaborative editing (M), with contribution tracking shown in three modes: (M-a) single-editor view where the original editor has not enabled collaborative editing, (M-b) the editor enables collaborative editing, making the draft available for others to refine, and (M-c) a subsequent collaborator edits the published draft, with contributions tracked across AI, the original editor, and the new collaborator.

editing is enabled, extends this to multi-author attribution, distinguishing the AI, the original editor, and each subsequent collaborator (Figure 1, F).

For each clip that corresponds to an AI-generated clip, we compute a retention score $R \in [0, 1]$:

$$R = w_{\text{text}} \cdot S_{\text{text}} + w_{\text{time}} \cdot S_{\text{time}} + w_{\text{pb}} \cdot S_{\text{pb}} + w_{\text{voice}} \cdot S_{\text{voice}} \quad (1)$$

S_{text} measures content preservation across three levels: lexical (normalized Levenshtein [26]), semantic (cosine similarity of BGE-M3 embeddings [10], which encode sentence meaning into dense vectors), and stylistic (cosine similarity of LUAR-MUD embeddings [37], which encode authorship patterns such as sentence structure and vocabulary choice).

$$S_{\text{text}} = w_{\text{lex}} \cdot S_{\text{lex}} + w_{\text{sem}} \cdot S_{\text{sem}} + w_{\text{sty}} \cdot S_{\text{sty}} \quad (2)$$

S_{time} uses Dynamic Time Warping [39], a sequence alignment method that measures how much a clip’s temporal placement was shifted in the video. S_{pb} and S_{voice} are binary indicators for playback type and delivery mode changes. Deleted clips receive $R = 0$; and inserted clips are attributed entirely to the human editor. The overall contribution score is:

$$C_{\text{AI}} = \frac{\sum_{i \in M} w_i \cdot R_i}{\sum_{i \in M} w_i + \sum_{j \in D} w_j + \sum_{k \in I} w_k}, \quad C_{\text{H}_1} = 1 - C_{\text{AI}} \quad (3)$$

where M, D, I are the sets of modified, deleted, and inserted clips; w_i is clip i ’s word count.

When a subsequent collaborator H_2 edits the H_1 -AI version, prior shares scale proportionally:

$$C_{H_2} = 1 - R', \quad C'_{H_1} = C_{H_1} \cdot R', \quad C'_{\text{AI}} = C_{\text{AI}} \cdot R' \quad (4)$$

This extends to any number of collaborators. We weight text and time equally ($w_{\text{text}} = w_{\text{time}} = 0.45$), with playback and voice at 0.05 each. Within the text dimension: $w_{\text{lex}} = 0.50$, $w_{\text{sem}} = 0.40$, $w_{\text{sty}} = 0.10$. These weights can be further calibrated as more editing data is collected on the platform.

4 Study Design

We conducted a within-subjects study with 30 novice describers to evaluate the GenAD-RefineAD pipeline and to test whether AI draft quality affects the editing experience.

4.1 Study Materials

4.1.1 Videos. We selected five YouTube videos, each approximately 2 minutes long, spanning three genres—instructional, entertainment, and educational—based on requested genres by the BLV community on YouDescribe [50]. Two instructional cooking videos, crispy fritters recipe¹ and potato balls recipe², were previously

¹<https://youtube.com/watch?v=cNj3aOTYdQQ>

²<https://youtube.com/watch?v=nqXz8hhAYGo>

Table 1: AI-generated draft characteristics across Baseline and GenAD conditions for all five study videos.

Video	Clips		Inline		Extended		Words	
	Base.	GenAD	Base.	GenAD	Base.	GenAD	Base.	GenAD
Crispy Fritters	24	24	17	23	7	1	307	298
Potato Balls	59	35	23	32	36	3	606	315
Tangled	23	15	16	15	7	0	416	354
Neuroscience	63	12	9	1	54	11	331	317
Origami	16	9	7	9	9	0	442	355

used in Yuksel et al. [52], providing a behavioral anchor to our from-scratch condition on the same content. To increase genre diversity and vary difficulty, we added three new videos: a clip from the animation *Tangled*³, a neuroscience lesson⁴, and an origami tutorial⁵.

Based on our consultants’ assessment, videos were categorized by difficulty. Cooking instructional and animated videos were classified as **low difficulty** due to their concrete visual content and sequential actions. Neuroscience and origami videos were classified as **high difficulty** because they require specialized technical terminology and involve abstract or spatially complex visual content.

4.1.2 Conditions. Each participant experienced three conditions across the five videos:

- **From scratch:** No AI draft; participants authored descriptions from a blank editor.
- **Baseline:** An unguided AI draft generated using a minimal prompt (“describe the scene”) without guidelines, context, or optimization.
- **GenAD:** The full pipeline output described in Section 3.1.

Table 1 summarizes the Baseline and GenAD drafts across all five videos. GenAD drafts consistently contained fewer clips overall and relied predominantly on inline playback, reflecting the optimization pass’s consolidation of descriptions into available speech gaps. Baseline drafts, generated without guideline and contextual awareness, produced more clips, a higher proportion of extended descriptions, and a generally higher total word count. This suggests that unguided prompting is less effective for abstract, domain-specific content where the model lacks technical context.

To further characterize the qualitative difference between conditions, our professional describer consultant independently reviewed sample drafts using a hierarchical 0–5 rubric assessing whether each description is true, useful, focused, polished, and well-timed. GenAD drafts, while imperfect, had most tracks reaching at least the usefulness threshold and were correctable in place. Baseline drafts, by contrast, contained too many clips with extensive overlap between tracks, making it impractical to evaluate individual descriptions. The consultant noted that Baseline drafts, particularly for the neuroscience video, would likely require deleting all tracks and starting fresh rather than editing incrementally.

The quantitative differences and the professional describer’s review confirm that participants encountered meaningfully different starting materials across the two AI conditions.

4.2 Participants

We recruited 30 participants (14 female, 16 male, ages 22–35) from a university campus. None had prior AD authoring experience, and about half (14) had never interacted with BLV users. This sample approximates the novice volunteer pool typical of crowd-sourced AD platforms but may not reflect the full diversity of that community in age, background, or ongoing motivation to contribute. Participants were compensated \$150 for their time.

4.3 Procedure

Prior to the session, participants received introductory materials including audio description guidelines and two 30-minute recordings of a professional describer authoring descriptions both from scratch and by editing an AI-generated draft in RefineAD.

The in-person session consisted of four phases: (1) an orientation covering the study goals, the RefineAD interface, and participant questions; (2) AD authoring for five videos, each assigned to one of the three conditions; (3) a NASA-TLX survey after each video to capture subjective workload; and (4) a post-session survey on overall experience. To keep sessions to a manageable length, participants used AI text-to-speech for voice delivery rather than recording their own voiceover.

Each participant completed one video from scratch, two with Baseline drafts, and two with GenAD drafts. The from-scratch condition was limited to low-difficulty cooking videos for two reasons. First, prior work established that novice describers require significant time to complete these videos from scratch, making higher-difficulty content impractical within a single session [52]. Extending this to high-difficulty content would have made sessions impractically long and introduced fatigue that could confound subsequent tasks. Second, using the same stimuli as [52] provides a behavioral reference point for comparing our from-scratch times against previous findings despite differences in interfaces and participant pools.

Participants were divided into four groups using a partial Latin square design. Condition assignments were counterbalanced so that each cooking video alternated between from-scratch and AI-assisted conditions across groups, while Videos 3–5 alternated between Baseline and GenAD across group pairs. Video order was randomized within each group to reduce order effects, and participants were not informed of the specific AI condition whenever possible (Appendix D). Each video was completed without interruption to ensure accurate time measurement; participants could rest between tasks after completing the post-video survey.

4.4 Measures

We measured task completion time and assessed perceived workload using the NASA-TLX [18] across conditions. Two-way repeated measures MANOVA was used to assess main effects and interactions between conditions with two levels of AI (Baseline and GenAD) and two difficulty levels (Low and High). For low difficulty videos where time and perceived workload were assessed under all three conditions (From-scratch, Baseline, and GenAD), one-way repeated measures MANOVA was used to analyze differences between the conditions. Sample sizes vary slightly across

³<https://youtube.com/watch?v=JXnAd8bofTo>

⁴<https://youtube.com/watch?v=6qS83wD29PY>

⁵<https://youtube.com/watch?v=djPgdlm6lMY>

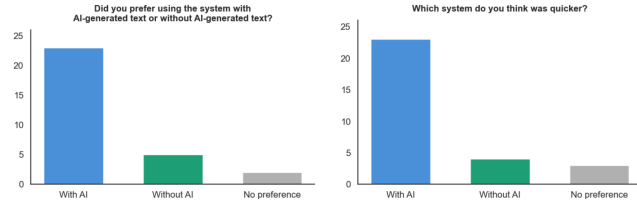


Figure 4: Participant preferences (left) and perceived speed (right). 76.7% preferred AI-assisted authoring and perceived it as quicker ($N = 30$).

comparisons due to missing timestamps or survey responses. Open-ended responses from post-study surveys were analyzed using inductive/deductive thematic consensus coding [5]; two researchers independently coded all responses and reconciled discrepancies through discussion.

5 Results

5.1 Quantitative Results

5.1.1 System Preference and Perceived Efficiency. Of the 30 participants, 23 (76.7%) preferred GenAD-RefineAD system, while 5 (16.7%) preferred authoring from scratch and 2 (6.7%) had no preference. Similarly, 23 (76.7%) perceived the system as quicker, 4 (13.3%) found from-scratch faster, and 3 (10.0%) saw no difference. Although preference and perceived speed largely aligned, a few participants diverged, suggesting that factors beyond efficiency, such as output quality or editing comfort, also influenced preference.

5.1.2 Task Completion Time. A two-way repeated measures MANOVA revealed a significant main effect for AI condition on task completion time (Pillai’s trace = 0.587, $p < 0.001$). No significant main effect of video difficulty (Pillai’s trace = 0.29, $p = 0.396$) and no AI condition $\times$ video difficulty interaction effect (Pillai’s trace < 0.01, $p = 0.985$) were found for task completion time. Participants using GenAD completed descriptions in approximately 14 minutes regardless of whether the video was low difficulty ($M = 13.98$, $SD = 9.94$) or high difficulty ($M = 13.85$, $SD = 11.22$), while participants editing on Baseline drafts required roughly half an hour for both low difficulty ($M = 29.99$, $SD = 18.16$) and high difficulty ($M = 31.92$, $SD = 19.23$) videos. A one-way repeated measures MANOVA conducted on only the low-difficulty videos, where all three conditions were available, found significant pairwise differences across all levels: From-scratch description ($M = 38.44$, $SD = 16.75$) took significantly longer than Baseline editing ($M = 29.99$, $SD = 18.16$) with $t(27) = 1.93$ and $p = 0.032$, which took significantly longer than GenAD editing ($M = 13.98$, $SD = 9.94$) with $t(28) = 2.02$, $p = 0.027$. Our from-scratch mean of 38.44 minutes is comparable to the 30.42 minutes reported by Yuksel et al. [52] on the same videos. Figure 5 shows the distribution of completion times across conditions.

5.1.3 Perceived Workload. The two-way repeated measures MANOVA also found a significant main effect of AI condition on NASA-TLX overall perceived workload (Pillai’s trace = 0.135, $p = 0.046$), with no interaction effect (Pillai’s trace = 0.009, $p = 0.625$) and no main effect for video difficulty (Pillai’s trace = 0.074, $p = 0.145$). GenAD was rated as less demanding than Baseline for both easy ($M = 29.22$ vs. $M = 37.24$) and hard ($M = 33.36$ vs. $M = 45.37$) videos. For

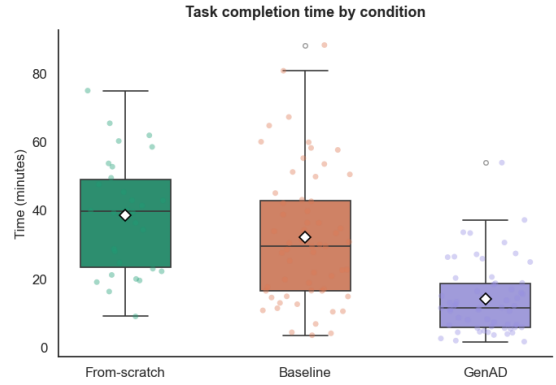


Figure 5: Task completion time by condition. GenAD ($M = 13.9$ min) was significantly faster than Baseline ($M = 31.0$ min) and From-scratch ($M = 38.4$ min). Diamonds indicate means.

Table 2: NASA-TLX overall workload by condition and video difficulty (0–100 scale; higher = greater demand).

Condition	Low Difficulty videos		High Difficulty videos	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
From-scratch	42.36	23.79	—	—
Baseline	37.24	25.21	45.37	26.97
GenAD	29.22	20.26	33.36	19.96

Note. Bold indicates the lowest workload per difficulty level.

easy videos across all three conditions, one-way repeated measures MANOVA found a significant difference in perceived effort only between from-scratch ($M = 42.36$) and GenAD ($M = 29.22$) conditions with pairwise $t(28) = 2.41$ and $p = 0.012$. Comparisons involving Baseline did not reach statistical significance, likely due to limited observed power, as participants completed only one from-scratch description. Notably, Baseline was rated as the most cognitively demanding condition for hard videos, suggesting that editing a lower-quality AI draft may impose additional cognitive load compared to editing a higher-quality draft. However, given the absence of a significant interaction effect, this pattern should be interpreted as exploratory rather than a confirmed finding. Table 2 reports workload scores by condition and difficulty.

5.1.4 How Participants Edited AI Drafts. We analyzed how participants used RefineAD’s editing features across the Baseline and GenAD conditions. Figure 6 shows the mean number of editing actions per description. Participants made fewer edits in the GenAD condition across all features. The largest difference was in deletions: Baseline descriptions averaged 11.3 deletes per session compared to 1.1 for GenAD, indicating that the unguided AI system produced excess or irrelevant clips that participants needed to remove. Text edits followed a similar pattern (5.6 vs. 2.4), as did playback type changes (2.1 vs. 0.4) and start time adjustments (2.5 vs. 1.2). Insertions were infrequent in both conditions (1.9 vs. 0.2). Overall, participants made fewer edits across all feature types in the GenAD condition, consistent with higher initial draft quality.

5.1.5 Contribution Score. Table 3 summarizes AI and human contribution by condition. GenAD drafts were largely retained ($M =$

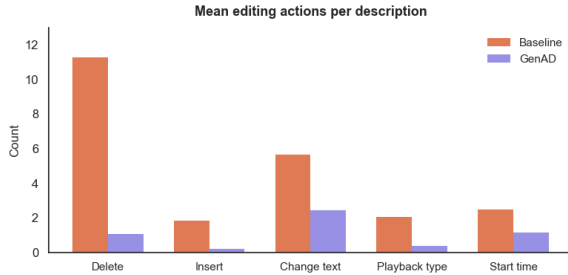


Figure 6: Mean editing actions per description by condition. GenAD drafts required fewer edits across all features, especially deletions.

Table 3: AI attribution by condition.

Condition	n	AI Attribution	Human Contribution
Baseline	57	67% $\pm$ 28%	33% $\pm$ 28%
GenAD	55	93% $\pm$ 9%	7% $\pm$ 9%

Note. Bold indicates the highest AI retention (lowest human editing effort).

Table 4: Correlation between human contribution and task completion time by condition. * $p < .01$, ** $p < .001$.

Condition	n	MDCI		Levenshtein	
		Pearson	Spearman	Pearson	Spearman
Overall	112	.70**	.71**	.66**	.68**
Baseline	57	.63**	.63**	.57**	.59**
GenAD	55	.40*	.59**	.35*	.53**

93%, $SD = 9\%$), with volunteers contributing only 7% on average. Baseline drafts required substantially more editing ($M = 67\%$, $SD = 28\%$), and the wider variance indicates greater individual differences in how volunteers engaged with lower-quality drafts.

We validated the contribution score by correlating our algorithmic results with task completion time (Section 5.1.2). We then compared this performance against a baseline of normalized Levenshtein distance calculated over the concatenated clip texts.

Both the MDCI and the Levenshtein method showed strong positive correlations with completion time ($\rho = .71$ and $.68$, $p < .001$), confirming that higher editing effort consistently required more time. Across all conditions, MDCI yielded consistently stronger correlations than Levenshtein. While the correlation was lower in the GenAD condition ($r = .40$ vs. $\rho = .59$), this was primarily due to the low variance in those scores ($SD = 9\%$); nevertheless, rank order associations remained moderate. As shown in Figure 7, human contribution tracks closely with the time spent by each participant.

To assess perceived fairness, we asked participants to rate how accurately the contribution score reflected their editing effort. Most found it reasonable: 14 (46.7%) rated it as "usually about accurate" and 11 (36.7%) as slightly off in either direction, 4 (13.3%) found it inconsistent, overestimating on some descriptions and underestimating on others, and only 1 (3.3%) felt it significantly overestimated their contribution. These ratings were collected using an earlier,

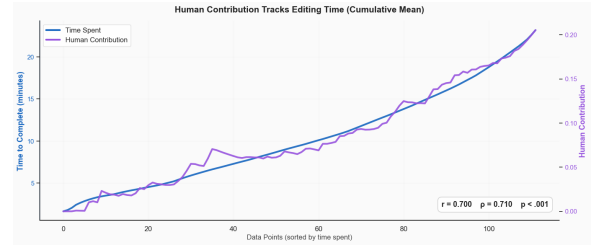


Figure 7: Human contribution tracks task completion time across participants ($\rho = .71$, $p < .001$), confirming the score’s alignment with observed editing behavior.

lexical-only version of the algorithm; the multi-dimensional formulation in Section 3.3 was developed in part to address cases where lexical distance alone misrepresented effort.

5.2 Qualitative Results

Three themes emerged from participants’ open-ended responses about their experience across conditions (Figure 8).

5.2.1 AI Drafts as Scaffolding for Description Authoring. Participants reported that AI-generated drafts reduced the effort required to begin the authoring process ($n = 16$). By providing immediate context and a starting point, drafts lowered the barrier to getting started: “Having an initial draft helped me focus on accuracy, timing, and clarity rather than starting from a blank page.” Beyond lowering the barrier to entry, drafts helped describers catch visual details they might otherwise overlook ($n = 8$). One participant remarked that with a draft, “it’s just easy and you don’t end up missing things in the video.” Others found that drafts aided their own understanding of the video content, citing “Text can help us understand the images and the video more clearly”, or surfaced information they had not identified on their own, “Provided text helps me to know some important info about the content.” This function is specific to AD authoring; unlike text editing where source material is static and re-readable, AD requires capturing transient visual information in real time. AI drafts effectively serve as a second pass over the video, catching what a single human viewing might miss. This scaffolding proved especially valuable for technical or abstract content, where describing from scratch felt daunting. For instance, with the neuroscience video, one participant noted that “it would have been challenging to write the descriptions without it,” since the content required “explaining visual diagrams for educational content.” Another observed that “it is very difficult even for an amateur person to describe this video.” A third noted that when content “relies heavily on the visuals, it’s a bit hard to explain through words alone.” For videos requiring domain terminology, spatial reasoning, or interpretation of abstract diagrams, the AI draft provided a foundation that most novice describers would have struggled to construct independently.

5.2.2 The Quality Threshold: When AI Drafts Help vs. Hinder. While participants valued AI assistance, its utility was conditional on draft quality ($n = 13$). Participants consistently framed this as a trade-off: high-quality drafts enabled a fluent editing experience ($n = 8$) that prioritized accuracy and clarity, while low-quality drafts imposed an editing burden that often exceeded the effort of writing from scratch ($n = 4$). As one participant noted: “Having a system with provided text is only efficient when it is accurate and to the point... I

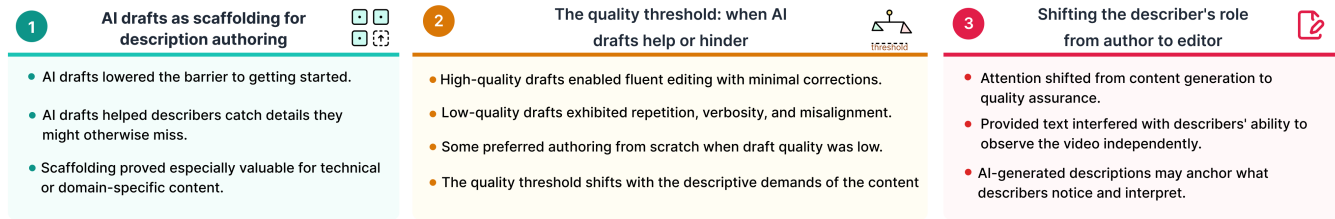


Figure 8: Three themes from participants' qualitative responses: AI drafts as scaffolding for description authoring, (2) a content-dependent quality threshold distinguishing when drafts help vs. hinder, and (3) AI drafts shifted the describer's role from author to editor, with both benefits and risks for novice skill development.

spent quite a lot time correcting it instead of having the time to describe it." Another echoed this conditionality: "When the suggested text is good quality, it only requires minor edits, allowing me to create higher quality AD without spending too much time." When the text met this bar, the experience was positive: "In almost all of the videos I found the text very helpful and I just needed to tweak it little bit."

When drafts fell short, participants identified consistent problems: unnecessary repetition ($n = 3$)—"unwanted repetition of neurons which was confusing", verbosity—"the description was too descriptive reporting unnecessary details", and incorrect temporal placement ($n = 2$)—"it was well generated text but it was playing too early." One participant noted that when corrections were minor the system was helpful, but "otherwise I would prefer to do it myself as I was able to focus on the accuracy and consistency of the video", suggesting that low-quality drafts added friction beyond the time cost alone.

This quality gap was most pronounced for high-difficulty, domain-specific content. In the neuroscience and origami videos, lower-quality drafts were described as "distracting," "frustrating," and "useless," often failing to explain spatial relationships or domain-specific logic. When participants received GenAD drafts for these same videos, the sentiment shifted, stating even with minor inaccuracies, these drafts were rated as "much better." For low-difficulty videos like cooking tutorials, participants raised fewer concerns about Baseline draft quality, suggesting that the quality threshold is not fixed but shifts with the descriptive demands of the content. Notably, participants arrived at these distinctions without knowledge of the study conditions, identifying quality differences purely through their editing experience. This content-dependent pattern suggests a higher tolerance for lower-quality drafts in familiar, sequential tasks (like cooking) but a much lower tolerance for technical or abstract material. This result is consistent with Yuksel et al.'s finding that even neutral-accuracy drafts are helpful for cooking videos [52], while extending it to show that these benefits do not necessarily generalize to more complex domains.

5.2.3 Shifting the describer's role from author to editor. Participants' responses also revealed how AI-generated drafts fundamentally changed the nature of the description task. With text already present, several describers reported that their attention shifted from content generation to quality assurance: checking accuracy, adjusting timing, and refining clarity. As one participant noted: "If the provided text is in a high quality, it would be easier to check the accuracy." Another framed their role as improving existing material: "I can change and improve by myself." This reframing of the task represents a qualitatively different cognitive demand. For novice describers with no prior AD experience, this editorial mode may

be more accessible than the generative mode. However, this role shift carries risks, particularly for novices. Some participants found that the presence of AI-generated text interfered with their ability to observe and interpret the video independently. One noted that "it can interrupt watching the video and make it harder to follow..." while another described cognitive overload from processing both video and text simultaneously: "Too much words in a short time with the same content of the pictures actually makes me dizzy." One participant who preferred the no-AI condition suggested that working without AI text avoided "bias for observation", implying that pre-existing descriptions may anchor what describers notice and how they interpret it. For novices who are still developing their observational and descriptive skills, this anchoring is especially concerning: if their primary experience of AD authoring is verifying AI text rather than engaging directly with visual content, they may not develop the perceptual habits that experienced describers rely on. These findings highlight a design tension in AI-assisted authoring tools: the editorial mode that makes the task accessible to newcomers may also limit their growth as independent describers.

6 Discussion

Our study demonstrates that AI draft quality determines whether human-AI collaboration helps or hinders audio description authoring. The scarcity of AD is fundamentally a supply-side problem: there are not enough describers producing descriptions for the volume of video content that exists and is requested by the BLV community. Our focus is therefore on the *authoring experience*, understanding what makes volunteer describers more efficient, less burdened, and more likely to produce and continue producing descriptions. We discuss what our findings mean for the design of AI-assisted authoring tools and how to support novice describers in building skills and sustaining contribution.

6.1 The Threshold Effect as a Design Principle

While draft quality is known to influence efficiency in machine translation and essay writing [13, 14, 17], our study provides the first evidence of a threshold effect in writing audio descriptions. The performance gap between GenAD and the Baseline, notable in both time and cognitive load, parallels Dhillon et al.'s finding ($N=131$) that low-level AI suggestions offered no benefit over unassisted writing, while high-level suggestions significantly improved productivity [14]. In both cases, the presence of AI assistance alone was not enough, its impact depended on whether the assistance met a minimum level of usefulness. In our data, this pattern is visible across both measures on low-difficulty videos: moving from no AI

to a low-quality draft reduced completion time by approximately 8 minutes and cognitive load by 5 points, whereas moving from a low-quality to a high-quality draft produced a substantially larger reduction of approximately 16 minutes and 8 points — consistent with a threshold rather than a linear relationship between draft quality and editing efficiency. By extending this pattern to AD, a task requiring complex visual judgment and temporal alignment, our results suggest that this quality floor extends beyond any single domain and reflects a broader pattern in human-AI collaboration.

This threshold is also content-dependent. For visually straightforward, sequential content like cooking videos, even lower-quality drafts may clear the threshold, a pattern consistent with Yuksel et al.’s finding that neutral-accuracy drafts still benefited novice describers [52]. A cross-study comparison on the same cooking videos reinforces this pattern directionally: Yuksel et al.’s HITL system produced descriptions in approximately 21 minutes [52], our Baseline condition required approximately 30 minutes, and GenAD reduced this to approximately 14 minutes, suggesting that prompting strategy and output quality matter more than AI presence alone. In contrast, for domain-specific content like neuroscience and origami, GenAD drafts met the threshold while Baseline drafts did not, as reflected in completion time differences and participants’ qualitative descriptions of editing difficulty — patterns participants identified without knowledge of condition assignments. The two-way MANOVA found no significant AI condition $\times$ difficulty interaction, so the content-dependency claim should be treated as a motivated hypothesis rather than a confirmed finding. We propose the threshold as a generalizable design principle: designers of AI-assisted authoring tools should not assume that any AI output provides a useful starting point, and should verify this threshold empirically for their target content and population before deployment.

6.2 Adapting to Content and Describer Needs

Given this content-dependent threshold, generation pipelines should invest prompting effort proportionally to content difficulty. GenAD currently applies a uniform pipeline to all videos; adapting generation investment to content complexity, adding domain-specific context for technical content while keeping prompts lighter for visually concrete content, could help drafts consistently clear the threshold across a wider range of videos. On the editing side, our consultants noted that experienced describers naturally adjust their approach by genre, prioritizing step-by-step accuracy for instructional content versus character and pacing for entertainment, but novices do not. Surfacing a video’s genre or functional goal in the editing interface could help novice describers orient their editing toward what matters most for the audience.

Describer characteristics represent another axis for adaptation. Several participants in our study were non-native English speakers and expressed preference for simpler vocabulary in drafts. Others described videos on familiar topics with greater confidence. These observations, while not formally tested as variables in our study, point toward a generation model that accounts for describer preferences, such as language complexity, desired level of detail, or topic familiarity, alongside content characteristics. A brief intake step that captures these preferences before draft generation could

reduce the mismatch between draft and describer intent, though this remains a design hypothesis to be validated in future work.

6.3 Designing for the Author-to-Editor Transition

The presence of AI drafts moves the describer away from active creation and toward a process of refinement: evaluating machine-generated text against visual content to ensure precise timing and polished language. Participants viewed this shift as beneficial, with 76.7% preferring AI-assisted authoring over manual writing. GenAD significantly reduced the cognitive load on users, demonstrating that the GenAD-RefineAD workflow successfully simplifies the descriptive process.

Because the editorial mode requires a tight edit-preview cycle, interface reliability is vital—particularly for novices, for whom a frustrating first experience can discourage continued participation [22]. Several participants reported technical friction during editing, most commonly needing to refresh the page for audio to play correctly after edits. While high-quality drafts provide a strong foundation, such disruptions, even when occasional, can create a bottleneck in an otherwise efficient editing workflow. Resolving playback reliability is therefore an immediate priority. Several participants also found the full set of editing features overwhelming on first use. We proactively addressed this by implementing a guided tutorial mode (Appendix E). This addition ensures that users are effectively onboarded before they begin the description task.

Designing for editing also means managing the risk that editors anchor on AI text rather than engaging independently with the video, a concern raised by participants and supported by prior work [2, 21]. RefineAD addresses this through its preview-first workflow and contribution tracking that keeps AI and human authorship visible. We plan to explore several directions to address this, including delaying AI text until after an unassisted viewing pass, presenting drafts with intentional gaps that prompt independent observation, and progressively reducing draft detail as describers gain experience [25].

All participants in our study were novice describers, who represent the primary contributor pool on volunteer-driven platforms like YouDescribe. Professional describers bring established mental models, domain expertise, and likely greater resistance to anchoring on AI text, their editing needs and the threshold at which drafts become productive may differ substantially. Prior work has studied these populations separately, designing authoring tools for sighted volunteers at varying expertise levels [8, 31] and for BLV creators [29]. An important next step is an adaptive system that tailors draft quality, interface complexity, and editing support to the describer’s expertise and access needs.

6.4 Contribution Tracking and Collaborative Agency

As AI-assisted authoring tools move from individual use to community-driven platforms, tracking who contributed what becomes essential. Contribution tracking makes visible the boundary between AI and human work across collaborators. The difference in AI retention between GenAD ($M = 93\%$, $SD = 9\%$) and Baseline ($M = 67\%$, $SD = 28\%$) illustrates how draft quality shapes editing patterns.

Over time, aggregating these patterns across content types could reveal where generation strategies consistently fall below the quality threshold, guiding targeted improvements to the pipeline.

On a deployed platform, this tracking serves multiple roles: giving volunteers credit, making the AI-human work distribution transparent, and enabling collaborative editing where subsequent describers can build on prior work with each contributor's changes attributed separately. For novices, viewing how others edited the same draft functions as informal training. The opt-in design preserves authorial agency, maintaining the sense of ownership that motivates contribution [23].

The current contribution weights ($w_{\text{text}} = w_{\text{time}} = 0.45$; $w_{\text{pb}} = w_{\text{voice}} = 0.05$) were set through expert consultation and validated against editing time. As the platform accumulates more editing data, we plan to recalibrate these weights by correlating algorithmic scores with describers' self-reported effort and perceived ownership, and by examining whether optimal weights differ across content types and describer expertise levels.

7 Limitations

GenAD relies on GPT-4o, whose computational cost scales with video volume and whose output characteristics shape the threshold we observed. Different models may produce drafts with different error profiles, shifting where the threshold falls and how editors interact with the text. Experimenting with a range of multimodal models, including smaller, open-source alternatives, would clarify whether the threshold is model-specific or generalizable, while also reducing cost for deployment at scale. Furthermore, our participant sample may not fully reflect the diversity of the broader crowd-sourced AD community. This study also focused on the authoring experience and did not evaluate the final descriptions. Prior work found that BLV users rated human-AI co-authored descriptions higher in quality and topic understanding than manually authored ones [52]; confirming whether GenAD drafts yield similar or stronger outcomes across varying content types is an immediate next step.

8 Conclusion

We present GenAD and RefineAD, a co-designed pipeline for audio description. In a within-subjects study ($N=30$), GenAD drafts cut completion time by more than half and significantly reduced cognitive load, while unguided drafts offered only minor time savings over writing from scratch—a difference participants recognized without knowing condition assignments. These results highlight a key design principle for human-AI authoring: effectiveness depends on output quality, not mere availability. Together, GenAD and RefineAD extend the human-AI workflow established by prior work into a quality-aware approach that points toward scalable, volunteer-driven audio description that lowers barriers while preserving essential human judgment.

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A GenAD Prompt Templates

This section contains the full prompt templates used in our system, including the system prompt with audio description guidelines, scene-level generation, optimization, and extended description filtering prompts.

A.1 System Prompt

The system prompt establishes the audio describer role and injects domain-specific guidelines that govern all downstream generation steps.

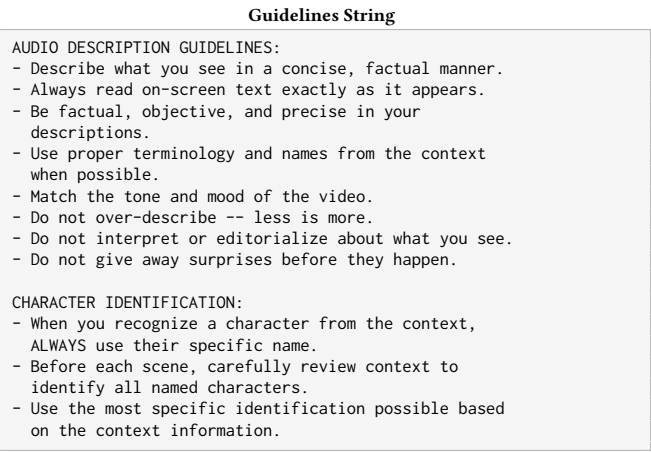


Figure 9: System prompt: the guidelines string is injected into the system message via the {guidelines} placeholder.

A.2 Scene-Level Generation Prompt

This prompt is issued once per scene. It receives the scene duration and narrative context, then requests structured JSON output for both on-screen text events and visual description events.

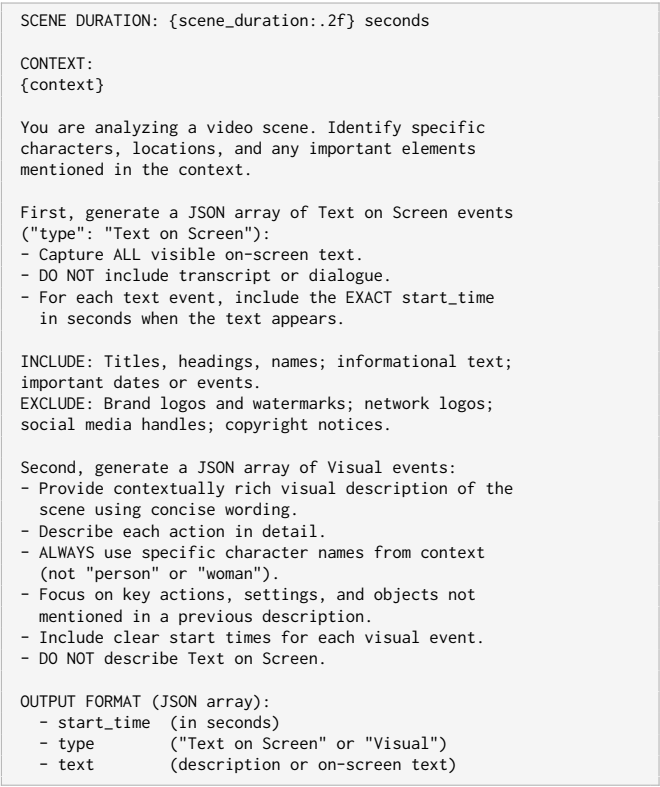


Figure 10: Prompt for generating scene-level descriptions and text events.

A.3 Inline Optimization Prompt

When the estimated speech duration of a generated description exceeds the available silence gap, the following prompt asks the model to condense the text to fit the timing constraint.

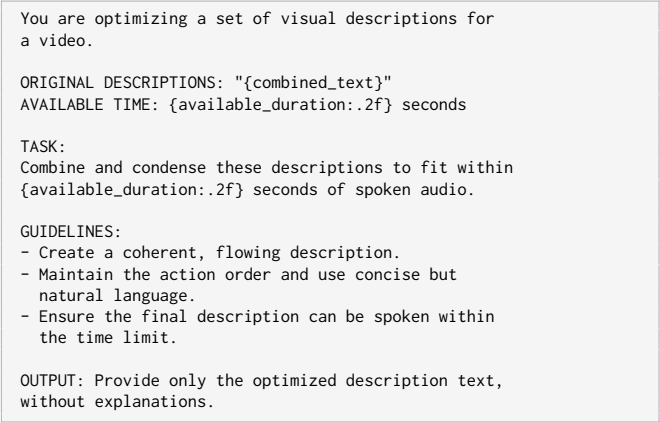


Figure 11: Prompt for inline description optimization.

A.4 Retry Optimization Prompt

If the first optimization attempt still exceeds the time budget (verified via TTS), this follow-up prompt provides the measured overshoot and requests further reduction.

```
You are optimizing visual descriptions for a video.

PREVIOUS ATTEMPT: "{optimized_text}"

This description takes {tts_duration:.2f} seconds to
speak, but only {available_duration:.2f} seconds are
available. Reduce by {tts_duration -
available_duration:.2f} seconds.

TASK:
Create a SHORTER version that fits within the time
limit.

GUIDELINES:
- Keep the most critical visual elements.
- Eliminate redundant details.
- Use concise but natural language.

OUTPUT: Provide only the shortened description,
nothing else.
```

Figure 12: Prompt for retrying a failed inline optimization.

A.5 Extended Description Filtering Prompt

For scenes where inline timing is insufficient, this prompt evaluates candidate descriptions against the transcript and cumulative context to decide whether an extended (pause-the-video) description is warranted.

```
You are an accessibility expert selecting ONE visual
description per scene to convert to audio description
for blind and low-vision users.

CONTEXT:
- AD should be minimal and only interrupt audio when
  necessary.
- Spoken transcript is the primary information source.
- Use AD only when critical visual information is
  missing from the audio.

INPUT:
CURRENT SCENE TRANSCRIPT: {transcript_text}
CUMULATIVE TRANSCRIPT:    {cumulative_transcript}
CUMULATIVE DESCRIPTION:   {previous_desc_text}
VISUAL DESCRIPTIONS TO EVALUATE: {clip['text']}

EVALUATION CRITERIA (mark necessary = true if any):
- Conveys visual details absent from the audio.
- Describes important silent actions or key events.
- Identifies characters or locations that are
  otherwise ambiguous.
- Introduces a novel or distinct visual element.
- Notes scene changes or unannounced time jumps.

OUTPUT FORMAT (JSON array, one item per description):
- id          (index of the description)
- necessary   (true or false)
- reason      (short explanation)
```

Figure 13: Prompt for filtering extended audio description candidates.

B AI-generated AD Preview Interface

Figure 14 shows the preview interface for AI-generated drafts. Once GenAD completes a draft, any platform user can preview the video with synchronized descriptions, rate individual clips, and begin collaborative editing to refine the draft, eliminating duplicate requests for the same video.

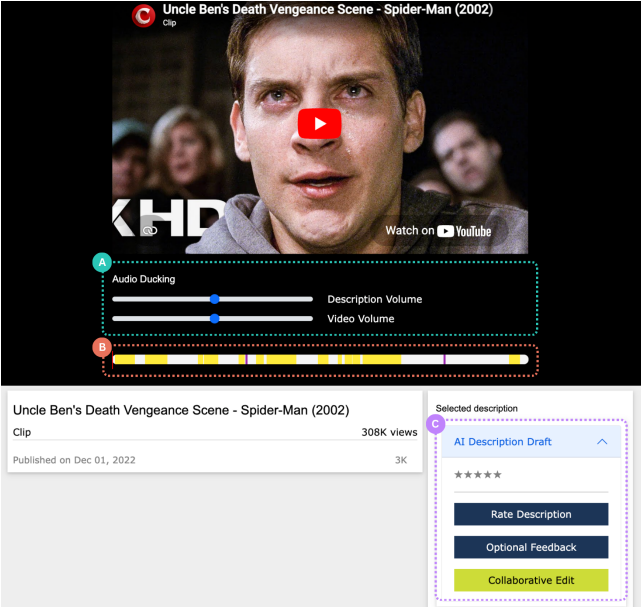
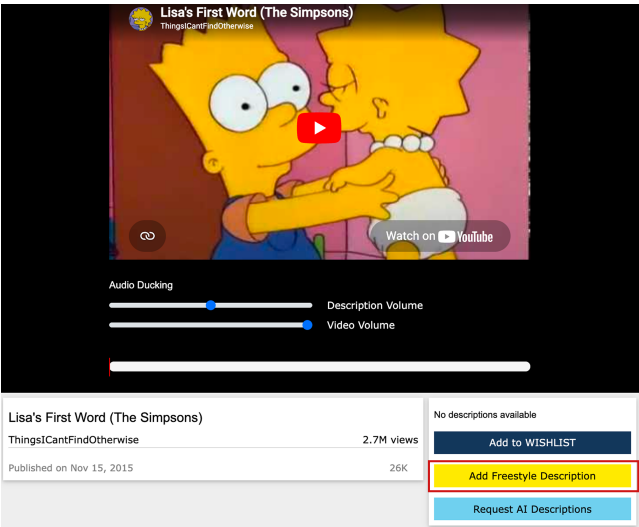


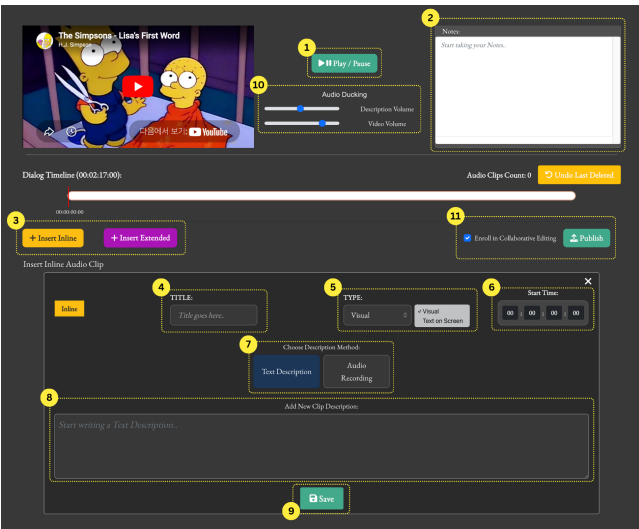
Figure 14: Video preview with AI-generated AD. Users watch the video with audio ducking controls (A) and view the timeline of inserted clips and dialogue (B; yellow = inline, purple = extended). From the selected description panel (C), users can rate the AI-generated description, provide optional feedback, or begin collaborative editing to refine the draft in RefineAD.

C From-Scratch Authoring Interface

Users can search for any YouTube video on the platform. When no descriptions are available, they can choose to add a freestyle description or request AI-generated descriptions. Selecting “Add Freestyle Description” (Figure 15a) opens the from-scratch authoring interface (Figure 15b), where describers compose and publish descriptions without AI assistance.



(a) Video landing page for an undescribed video. Selecting “Add Freestyle Description” (highlighted) opens the authoring interface shown in (b).



(b) From-scratch authoring interface. Describers preview the video with play/pause controls (1) and take notes (2). They then insert new inline or extended descriptions (3; yellow = inline, purple = extended), add a title (4), select the description type (5), adjust the start time (6), choose a description method (7), write the description text (8), and save changes (9). Audio ducking is controlled via (10), and the video can be published with optional collaborative editing (11).

Figure 15: From-scratch authoring workflow. (a) Video landing page showing available actions when no descriptions exist. Selecting “Add Freestyle Description” (highlighted) opens the authoring interface in (b). (b) The editing interface where describers compose and publish descriptions without AI assistance.

D Study Interface and Condition Assignment

Table 5 shows the full condition assignment across groups. Figure 16 shows the anonymized condition labels presented to participants.

Table 5: Condition assignment across four groups. Video order was randomized within each group.

	Video 1 <i>Crispy Fritters</i>	Video 2 <i>Potato Balls</i>	Video 3 <i>Tangled</i>	Video 4 <i>Neuroscience</i>	Video 5 <i>Origami</i>
Group 1	From scratch	GenAD	Baseline	GenAD	Baseline
Group 2	Baseline	From scratch	GenAD	Baseline	GenAD
Group 3	GenAD	From scratch	Baseline	GenAD	Baseline
Group 4	Baseline	From scratch	GenAD	Baseline	GenAD

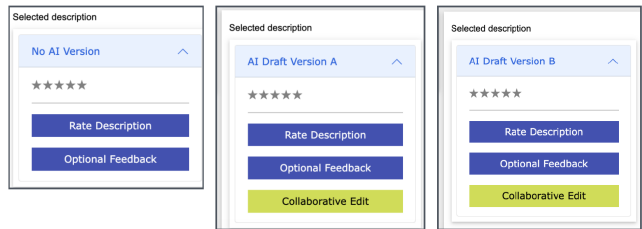


Figure 16: Condition labels presented to participants. The two AI conditions were labeled *AI Draft Version A* and *AI Draft Version B* to avoid signaling differences in generation approach.

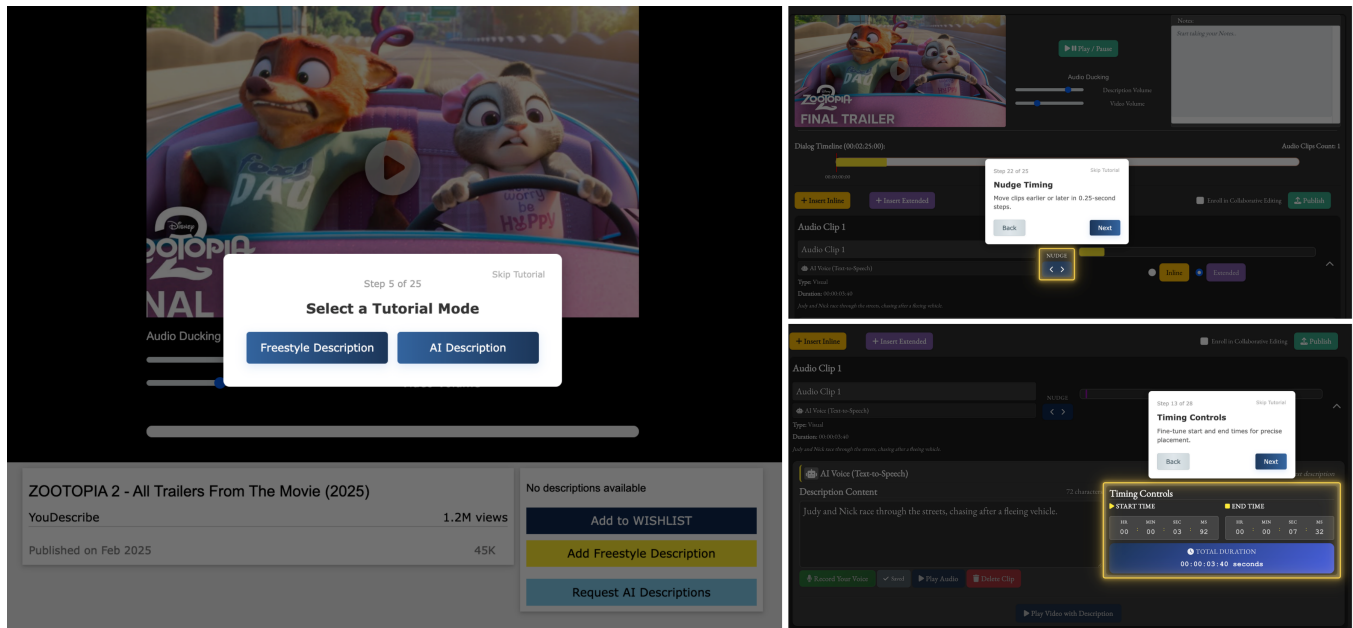


Figure 17: Interactive onboarding tutorial. A spotlight overlay guides users step by step through the interface, requiring specific interactions before advancing. At an intermediate stage, the tutorial branches into manual authoring or AI draft refinement paths.

E Interactive Onboarding Tutorial

Based on participant feedback that the full set of editing features felt overwhelming on first use, we implemented an interactive tutorial mode (Figure 17) to onboard new users before they begin describing.